

Circuit Breakers

7.1 Introduction

The protective switch gear is designed based on fault Analysis

- Circuit breaker is designed based on most severe fault at its location. (3 - ϕ Fault).
- If there is a fault on any part of the power system, the magnitude of fault current is very high. (Approximately 2 to 10 times the full load current).
- The fault current is not just limited to faulty section of the system but even extended to each and every healthy sections of the system.
- The relay is a device which senses the abnormal operation of power system and issue signal to corresponding circuit breaker and circuit breaker disconnect faulty section from healthy system.

7.2 Types of protective switches used in the system

- Fuse
- Load interrupter
- Line isolator
- Circuit breaker

(i) Fuse

- Operates only under abnormal conditions
- Whenever heat energy developed by the fuse is more than its thermal limit, then the fuse will melt down or blown out, fault element can be disconnected.
- Fuse is a primary protective device in the distribution system
- In substation, fuse is a backup protective device.
- The fusing current, $I \propto d^{3/2}$, 'd' is the diameter of the wire.
- The fuse element is generally made up material having the following characteristic.

(a) Low melting point

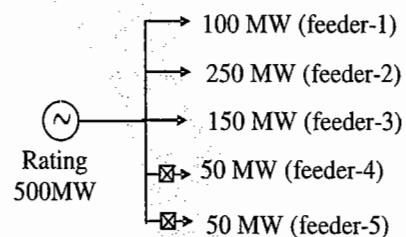
Eg: tin, lead, zinc

- Free from deterioration due to oxidation,
- low cost

(ii) Load interrupter

- This will be operated only at full load conditions.
- Fuse is operates at abnormal conditions (itself).
- Load interrupter is used for energy management, load shedding/curtailment
- Breaking capacity of load interrupter = full load capacity.

Eg:

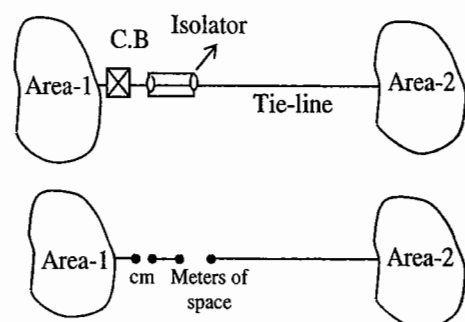


$\Rightarrow$ Total load = 600 MW

The above diagram represents over load on the system i.e 100 MW. The load interrupter operates and disconnects feeder-4 and feeder-5

(iii) Line Isolator

- It will be operated only under no-load condition.
- The necessity of Line Isolator is to provide more clearance.
- Sequence of operation is: First load interrupter has to operate and next Line Isolator.
- Breaking capacity of Line Isolator is zero





(iv) Circuit breaker

- It will be operates under all conditions i.e., at no-load or at full load or at abnormal condition.
- Fuse and load interrupter can be replaced by circuit breaker.
- Breaking capacity of C.B is three phase short circuit MVA at breaker location.
- C.B is a protective switch and which will make the circuit and break the circuit during normal and abnormal conditions respectively.

7.3 Purpose of circuit breakers

The main purpose of circuit breakers is to:

- Switch load currents
- Make on to a fault
- Break normal and fault current
- Carry fault current with out damage till the interruption.

The important characteristics from protection point of view are:

- It should open main contact as quickly as possible once it receives trip signal from relay
- The capacity of circuit that the main contact is capable of interrupting

The first characteristics referred to as the tripping time of circuit breaker (CB open time), which is expressed in cycles.

The opening time of modern high-speed circuit breaker is $1\frac{1}{2}$ cycle to 8 cycles.

Opening time:

It is the time between the instant of application of tripping power to the instant of separation of main contacts.

Arcing time:

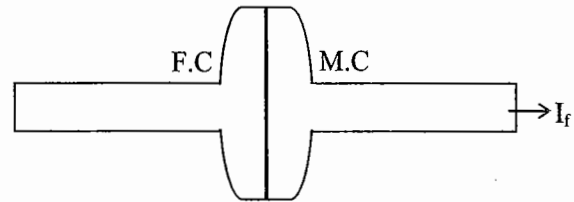
It is the time between the instant of separation of main CB contacts to the instant of Arc extinction.

CB operating time = opening time + Arcing time

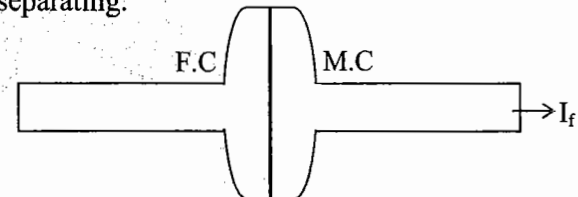
7.4 Arc phenomenon Circuit breaker

7.4.1 ARC Initiation

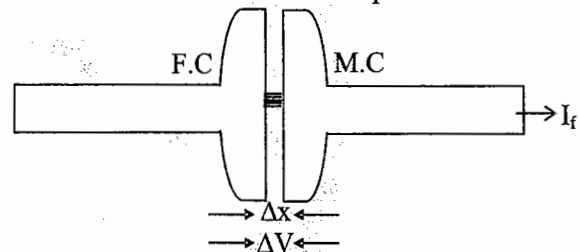
At $t = 0^-$: There is No trip signal to C.B From relaying system.



At $t = 0$: There is a trip signal to C.B from Relaying system & CB contacts are started separating.



At $t = 0^+$: CB contacts little separated.



Ex: 132 kV C.B

Δx - Distance between fixed contact and moving contact (which is very less $\Delta x = 0.1 \mu m$)

ΔV - Voltage across contact spacing (which is very less compared to operating voltage $\Delta V = 100 V$)

Electric field intensity between the contacts,

$$E = \frac{\Delta V}{\Delta x}, E = \frac{100 V}{0.1 \mu m} = 1000 kV / cm$$

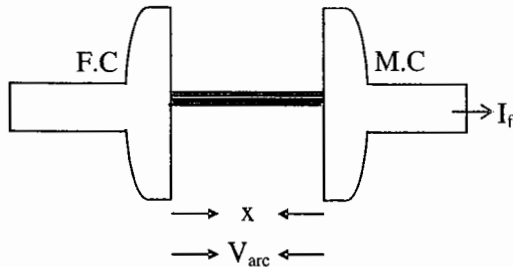
'E' is very high and it is definitely greater than the breakdown strength of the medium between contacts. So gap breakdowns and arc is initiated.

Conclusion: arc is initiated at the instant of contact separation due to high field gradient (or) field ionization.

7.4.2 ARC Maintenance

EX: $1\frac{1}{2}$ cycle C.B; Open time = 30 ms.

At $t = 30$ ms: Contacts are fully separated.



X – Contacts Spacing (it is in c.m)

$$\therefore x = 1 \text{ cm}$$

V_{arc} – Arc voltage (it is 2 to 5% of system voltage)

$$E = \frac{V_{\text{arc}}}{x} = \frac{0.03 \times 132 \text{ kV}}{\sqrt{3}} \times \frac{1}{1 \text{ cm}} \\ = 2.286 \text{ kV/cm.}$$

So field gradient is very less compare to the dielectric strength of the medium between the contacts.

Conclusion:

- The Arc is maintained not because of field gradient
- It is maintained because of thermal ionization due to high amount of heat produced during Arcing process.

7.4.3 Properties of the Arc:

Arc is column of ionized gasses.

- Conductance of their Arc
 - $\propto$ Number of free e^- generated during ionization.
 - $\propto$ Cross section areas of the Arc
 - $\propto \frac{1}{\text{length of the Arc}}$
- The path of the Arc is purely resistive and it is negative temperature coefficient Temperature increases $\Rightarrow R_{\text{arc}}$ decreases

$\Rightarrow$ Conductance of the Arc increases.

- The resistance of the Arc is highly non-linear.

7.5 ARC interruption

There are two methods of arc interruption:

- High Resistance method
- Current zero interruption method.

7.5.1 High Resistance Method

In this method, ARC resistance is increased so as to reduce the current to a value which is insufficient to maintain the Arc.

$$\text{Heat energy} = I^2 R_{\text{Arc}} t$$

$$I = \frac{V_{\text{Arc}}}{R_{\text{Arc}}}$$

$$\text{Heat energy} = \frac{V_{\text{Arc}}^2}{R_{\text{Arc}}} \times t$$

$$\text{Heat energy} \propto \frac{1}{R_{\text{Arc}}}$$

$$R_{\text{Arc}} = \frac{\rho l}{A}; \rho = \text{Resistivity of the Arc}$$

l = Length of the Arc

A = Cross section area of the Arc

The Arc resistance can be increased.

- Lengthening the Arc
 - Constraining the Arc
 - Cooling the Arc
 - Reducing cross section area of the Arc.
- When current is interrupted the energy associated with its magnetic field appears in the form of electro static energy. So a high voltage appears across the breaker contacts.
 - If the voltage across the breaker contact is very high and it is greater than the with stand capacity of the gap then gap will breakdown and Arc will restrike.
 - Therefore this method of Arc interruption is not suitable for high current interruption.
 - This method is employed for low power A.C & D.C circuit breakers.

7.5.2 Current zero interruption

- This method is applicable only in case of A.C circuit breakers.
- For a 50Hz system, the current wave passes through natural zero 100 times per second. This feature of Arc is utilized for A.C interruption.
- In this method the current is not allowed to rise again after zero current occurs.
- The current is not interrupted other than zero current instant, a high transient voltage appearing across the breaker contacts which restrike the Arc again.

There are two theories which are explaining current zero interruption method

- Recovery rate theory. (Slepain's theory)
- Energy balance theory. (Cassies theory).

Recovery rate theory (Voltage race theory)

- Arc is a column of ionized gasses.
- The ionization is minimum at current zero instant. So to extinguish the arc, ions and electrons removed from the gap immediately after the current reaches to a natural zero.
- Ions and electrons can be removed either by recombining them into neutral molecule or by sweeping them by inserting an insulating medium between the contacts.
- The Arc is interrupted if rate of removal of ions from the gap is greater than that of rate of ionization.
- Rate of ionization is depends on restriking voltage.
- So in this method, the rate at which dielectric strength recovered is compared with rate of rise of restriking voltage.
- If dielectric strength increases more rapidly than restriking voltage the arc is extinguished.
- If restriking voltage raises more rapidly than the dielectric strength, the ionization persist and breakdown of the gap occurs, resulting in an arc for another half cycle.

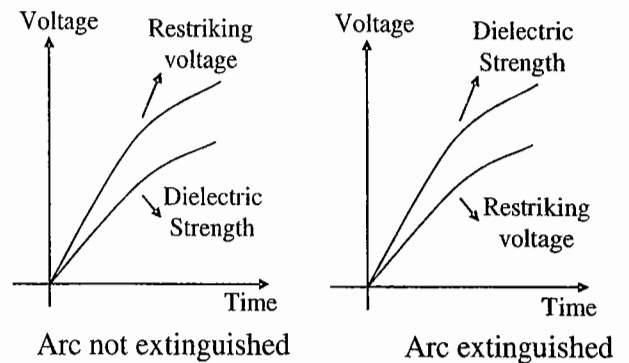


Fig. 7.1: Recovery rate theory

Energy balance theory

- The C.B contacts are about to open its resistance is zero and heat generation is zero.
- When contacts are fully opened and arc is extinguished resistance is infinity and heat generation is again zero.
- Between these two instant heat generation goes to a maximum value.

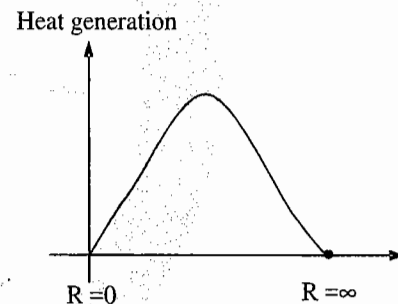


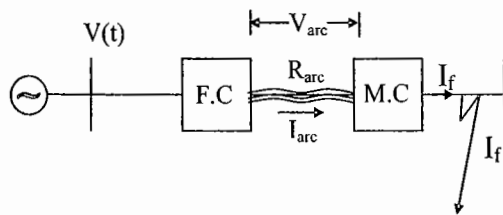
Fig.7.2 Energy balance theory

- The circuit breaker is designed to remove this generated heat as early as possible by cooling the gap, giving a blast of air or flow of oil at high velocity and pressure.
- If the rate of heat generation is less than the rate of heat dissipation, the arc is extinguished.
- If the rate of heat generation is more than the rate of heat dissipation. The space break down and resulting in an arc for another half cycle.

7.6 Terms regarding Arc phenomena

(i) Arc voltage (V_{arc})

Voltage across circuit breaker during arcing period is known as arc voltage



- Arc will be modeled as a non linear resistor, where R_{arc} is variable according to temperature.

$$V_{arc} = I_{arc} R_{arc}$$

As I_{arc} increases, temperature increases and R_{arc} decrease. So $V_{arc} \approx \text{constant}$

- V_{arc} , I_{arc} are in phase
- Arc power factor is unity but fault power factor is almost zero lagging.

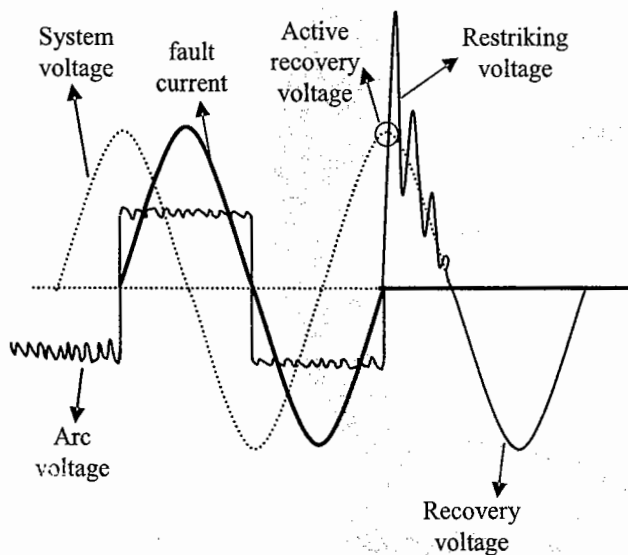


Fig. 7.3: Arc Phenomenon

(ii) Active Recovery Voltage

The instantaneous voltage at the instant of Arc extinction is called active recovery voltage.

Active recovery voltage depends on following factors

- The power factor of the fault current.
 - The Armature reactions synchronous machine
 - The circuit conditions [Type of fault, System grounding]
- If operating voltage of the system is 132 kV and all factors are neglected then

$$V_{arv} = \frac{132}{\sqrt{3}} \times \sqrt{2} \text{ kV}$$

V_{arv} = Maximum value of phase voltage of system (V_m)

- To account for power factor of the fault ' V_m ' should be multiplied by a factor $\sin \phi$
Where ' ϕ ' is the power factor angle of fault current.
- To account for effect of armature reaction ' V_m ' should be multiplied by a factor K_1
 $K_1 \leq 1$.
- To account for circuit conditions ' V_m ' should be multiplied by a factor ' K_2 '

K_2 is called first pole clearing factor

$$K_2 = \frac{\text{Voltage across first pole which is cleared the fault}}{\text{Voltage across C.B after all poles are cleared the fault}}$$

Type of Fault	LG	LL	LLG	LLL	LLLG
Grounding					
Grounded system	$K_2 = 1$	$K_2 = 1.5$	$K_2 = 1$	$K_2 = 1.5$	$K_2 = 1$
Ungrounded system		$K_2 = 1.5$	$K_2 = 1.5$	$K_2 = 1.5$	$K_2 = 1.5$

$$V_{arv} = K_1 K_2 V_m \sin \phi$$

Note:

To calculate line value of active recovery voltage the above expression should be multiplied by a factor $\sqrt{3}$.

Ex: For a 132 kV system

$$V_m = \frac{132 \text{ kV}}{\sqrt{3}} \times \sqrt{2}$$

(iii) Restriking voltage

The transient voltage across breaker contacts at the instant of arc extinction is known as restriking voltage.

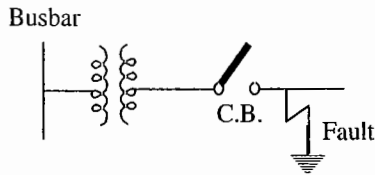


Fig 7.4: Fault on terminals of a circuit breaker

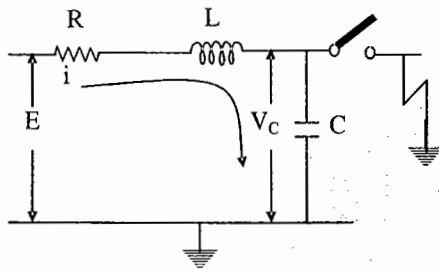


Fig 7.5: Equivalent electrical circuit for analysis of restriking voltage

$$L \frac{di}{dt} + \frac{1}{C} \int i dt = E$$

Where E is the system voltage at the instant of arc interruption (also called active recovery voltage). As the transient oscillation is a fast phenomenon, E can be regarded as a constant for a short duration.

Restriking voltage = Voltage across the Circuit breaker at the instant of arc extinction

= Voltage across the Capacitor at the instant of arc extinction

$$i = \frac{dq}{dt} = \frac{d(CV_c)}{dt}$$

Where V_c = voltage across the capacitor.

$$\text{Therefore, } \frac{di}{dt} = \frac{d^2(CV_c)}{dt^2} = C \frac{d^2V_c}{dt^2}$$

$$\int \frac{idt}{C} = \frac{q}{C} = V_c$$

Substituting these values in the equation given above, we get

$$LC \frac{d^2V_c}{dt^2} + V_c = E$$

Taking Laplace Transform on both sides of the equation, we get

$$LC s^2 V_c(s) + V_c(s) = \frac{E}{s}$$

Where $v_c(S)$ is the Laplace Transform of v_c .

Other terms are zero as initially $q = 0$ at $t = 0$,

$$\text{or } V_c(s)[LCs^2 + 1] = \frac{E}{s}$$

$$\text{or } V_c(s) = \frac{E}{s[LCs^2 + 1]} = \frac{E}{LCs \left(s^2 + \frac{1}{LC} \right)}$$

$$\omega_n = \frac{1}{\sqrt{LC}}, \text{ therefore, } \frac{1}{LC} = \omega_n^2$$

$$\text{or } V_c(s) = \frac{\omega_n^2 E}{s(s^2 + \omega_n^2)} = \frac{\omega_n E}{s} \left(\frac{\omega_n}{s^2 + \omega_n^2} \right)$$

Taking the inverse Laplace, we get

$$\begin{aligned} V_c(t) &= \omega_n E \int_0^t \sin \omega_n t \\ &= \omega_n E \left[\frac{-\cos \omega_n t}{\omega_n} \right] \end{aligned}$$

As $V_c(t) = 0$ at $t = 0$, constant = 0.

$$V_c(t) = E(1 - \cos \omega_n t)$$

= Restriking voltage

- The maximum value of restriking voltage
= $2 E_{\text{peak}}$
= $2 \times$ peak value of the system voltage
- The Rate of Rise of Restriking Voltage (RRRV)

$$\begin{aligned} &= \frac{d}{dt} [E(1 - \cos \omega_n t)] \\ &= \omega_n E \sin \omega_n t \end{aligned}$$

- The maximum value of RRRV = $\omega_n E$
= $\omega_n E_{\text{peak}}$

- Frequency of natural oscillation,

$$\omega_n = \frac{1}{\sqrt{LC}} \text{ rad/sec}$$

$$f_n = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$

7.7 Current chopping

When interrupting a low inductive current (shunt reactor or magnetizing current of transformer), the current become abruptly zero well before natural zero instant. This phenomenon known as current chopping.

Due to current chopping the voltage across circuit breaker under goes to huge oscillation and may results in restriking.

During current chopping,

$$\frac{1}{2}Li^2 = \frac{1}{2}cV^2 \Rightarrow v = i\sqrt{\frac{L}{C}}$$

$$\text{Prospective voltage } v = i\sqrt{\frac{L_{eq}}{C_{eq}}}$$

- A current chopping phenomenon is very severe during the interruption of low magnetizing current.
- A current chopping phenomenon is very severe in air blast circuit breaker.

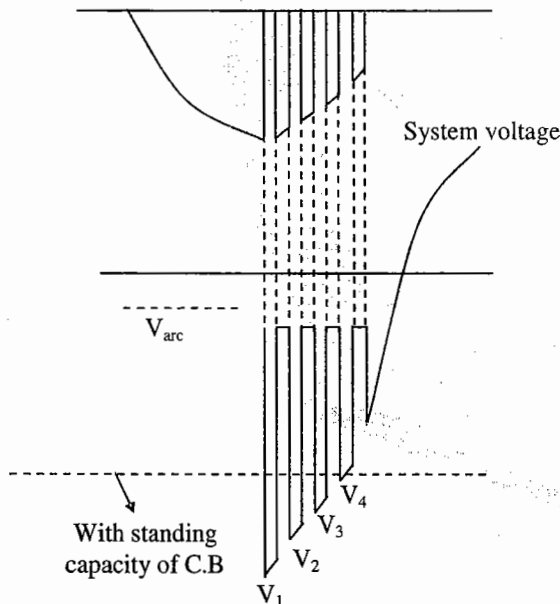


Fig. 7.6: Current Chopping

- If prospective voltage > with standing capacity of circuit breaker then gap will restriking.
- Arc restriking is successfully done up to the point where the prospective voltage become less than withstanding capacity of circuit breaker.

7.8 Resistance Switching

The resistance switching is employed to reduce the effect of restriking voltage and current chopping.

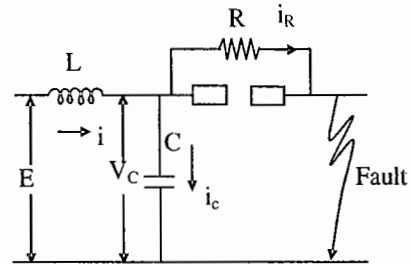


Fig. 7.7: Circuit for analysis of resistance switching

The voltage equation is given by

$$L\frac{di}{dt} + \frac{1}{C}\int i_c dt = E \text{ and } i = i_c + i_R$$

Therefore, the above equation become

$$L\frac{d(i_c + i_R)}{dt} + V_C = E$$

$$L\frac{di_c}{dt} + L\frac{di_R}{dt} + V_C = E$$

$$\text{Or } i_c = \frac{dq}{dt} = \frac{d(CV_C)}{dt}$$

$$\text{Therefore, } \frac{di_c}{dt} = \frac{d^2(CV_C)}{dt^2} = C\frac{d^2V_C}{dt^2}$$

$$\frac{di_R}{dt} = \frac{d(V_C/R)}{dt} = \frac{1}{R}\frac{dV_C}{dt}$$

Substituting these values in the main equation. We get

$$LC\frac{d^2V_C}{dt^2} + \frac{L}{R}\frac{dV_C}{dt} + V_C = E$$

Taking Laplace Transform, we get

$$LCs^2V_C(s) + \frac{L}{R}sV_C(s) + V_C(s) = \frac{E}{s}$$

Other terms are zero, as $V_C = 0$ at $t = 0$

$$\text{or } LCV_C(s)\left[s^2 + \frac{1}{RC}s + \frac{1}{LC}\right] = \frac{E}{s}$$

$$\text{or } V_C(s) = \frac{E}{sLC\left[s^2 + \frac{1}{RC}s + \frac{1}{LC}\right]}$$



For no transient oscillation, all the roots of the equation should be real. One root is zero, i.e. $s = 0$ which is real. For the other two roots to be real, the roots of the quadratic equation in the denominator should be real. For this, the following condition should be satisfied.

$$\left[\left(\frac{1}{2RC} \right)^2 - \frac{1}{LC} \right] \geq 0 \text{ or } \frac{1}{4R^2C^2} \geq \frac{1}{LC}$$

$$\text{or } \frac{4}{LC} \leq \frac{1}{R^2C^2} \text{ or } R^2 \leq \frac{LC}{4C^2}$$

$$\text{or } R^2 \leq \frac{1}{4} \cdot \frac{L}{C} \text{ or } R \leq \frac{1}{2} \sqrt{\frac{L}{C}}$$

Therefore, if the value of the resistance connected across the contacts of the circuit breaker is equal to or less than $\frac{1}{2} \sqrt{\frac{L}{C}}$, there will be no oscillations.

System damping	Value of R	Envelope of Restriking Voltage
Critical damping	$R = \frac{1}{2} \sqrt{\frac{L}{C}}$	
Over damping	$R < \frac{1}{2} \sqrt{\frac{L}{C}}$	
Under damping	$R > \frac{1}{2} \sqrt{\frac{L}{C}}$	
Un-damping	$R = \infty$	

The frequency of damped oscillation is given by

$$f = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{1}{4C^2R^2}}$$

7.9 Ratings of circuit breakers

1. Breaking capacity of circuit breaker
2. Making capacity of circuit breaker
3. Short time current rating:

This is based on thermal and mechanical considerations. The circuit breaker must be capable of carrying short circuit current for a short period without damaging its contacts.

4. Rated voltage $\rightarrow$ L - L rms voltage, generally more than the nominal system voltage.
5. Rated current or continuous current rating: It is rms value of current carried by CB under normal operating condition
6. Rated frequency $\rightarrow$ System frequency
7. No. of operation.

No. of operation is specified on name plate of CB after which the CB requires maintenance.

One operation = One break + one make

8. Operating duty

(i) Circuit breaker not used for auto reclosing.

B - 3 - MB - 3 - MB

Where, B - breaking operation,

3 - Duration in minute

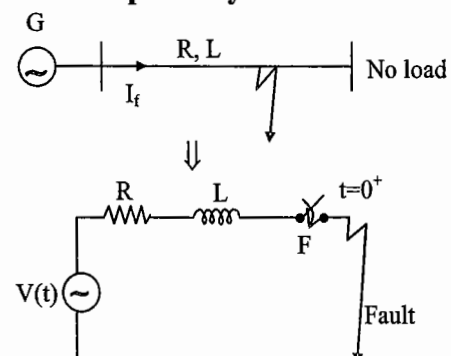
M - Making operation

(ii) Circuit breaker with auto reclosures

B - D_t - MB

Where, D_t - Det time of circuit breaker in cycles (approximately 300 ms).

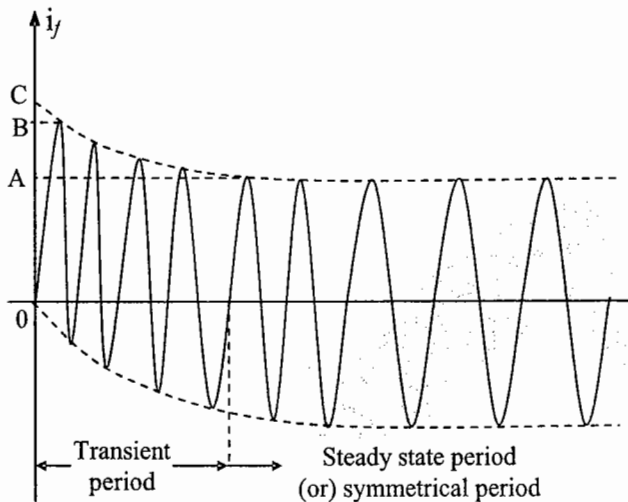
Ex: Fault on power system



$$i_f = k_1 e^{-\frac{Rt}{L}} + \frac{V_m}{|Z|} \sin(\omega t + \alpha - \phi)$$

$$\phi = \tan^{-1} \left(\frac{\omega L}{R} \right)$$

$$|Z| = \sqrt{R^2 + (\omega L)^2}$$



7.9.1 Symmetrical breaking capacity

It is the rms value of the ac component of the fault current that the circuit breaker is capable of breaking.

- Symmetrical breaking current, $I_{sy} = \frac{OA}{\sqrt{2}}$
- I_{sy} is the fault current corresponds to 3- ϕ fault at breaker terminal

$$I_{sy} = \frac{V_{pre}}{Z_{th}}$$

- Symmetrical breaking capacity

$$= \sqrt{3} \times \text{rated system voltage} \times \text{symmetrical breaking current}$$

$$= I_{sy} (p.u.) \times \text{Base MVA}$$

$$= |S.C. MVA|_{3-\phi \text{ fault}}$$

7.9.2 Asymmetrical breaking capacity

It is the rms value of the total current comprising of both ac and dc components of the fault current that the circuit breaker can break.

Asymmetrical current (For First cycle)

$$= \left(\frac{OA}{\sqrt{2}} \right)^2 + (AB)^2$$

$$= x \times I_{sy}$$

$$x = 1.6 \text{ for first cycle}$$

Asymmetrical breaking capacity:

$$= \sqrt{3} \times \text{rated system voltage} \times \text{Asymmetrical breaking current}$$

$$= x \times \text{symmetrical breaking capacity}$$

Note: Default is always symmetrical breaking capacity.

7.9.3 Making Capacity

The possibility of a circuit breaker to be closed on short circuit is also considered. The rated making current is defined as the peak value of the current (including the dc component) in the first cycle at which a circuit breaker can be closed onto a short circuit.

- The capacity of a circuit breaker to be closed onto a short circuit depends upon its ability to withstand the effects of electromagnetic forces.

Making current = the peak value of fault current corresponds to first cycle

$$= OC$$

$$= \sqrt{2} \times 1.8 \times \text{symmetrical breaking current}$$

$$= 2.55 \times \text{symmetrical breaking current}$$

Making capacity

$$= 2.55 \times \text{symmetrical breaking capacity}$$

7.10. Classification of Circuit Breakers

Classification is based on Arc quenching medium (or) insulating medium used between contacts

- Air break circuit breaker
- Air blast circuit breaker
- Oil circuit breaker
- SF₆ gas circuit breaker
- Vacuum circuit breaker

In circuit breakers, the solid insulating materials will not be used for arc quenching purpose



7.10.1. Air break circuit breaker

- Also called AB switch.
- Atmospheric air available at room temperature and normal pressure will be used as Arc quenching medium
- Dielectric strength of air = 30 kV/cm (peak)
- There is no external action taken for the Arc interruption in the air break circuit breaker. So Arc interruption process is poor
- This circuit breaker used for operating voltages (400 V to 1000 V)

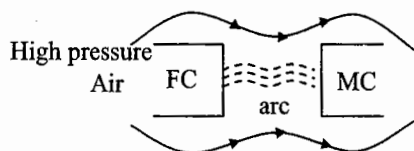
7.10.2. Air blast circuit breaker

High pressure air will be used as arc quenching medium

- High pressure air will be blasted into the Arcing chamber for Arc interruption.
- High pressure air will have good cooling properties
- The high pressure air will be maintained up to condensation point where the gaseous material will be converted into liquid material.
- Air pressure is maintained at 20 – 30 kg/cm²
- Operating voltage levels of Air blast circuit breakers is 12 kV to 525 kV
- According to blasting direction of air there are three types of Air blast circuit breakers.

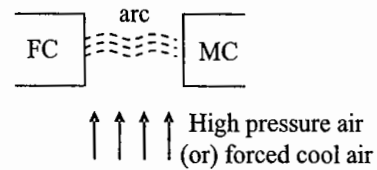
- Axial blast
- Cross blast
- radial blast

(i) Axial blast circuit breaker:



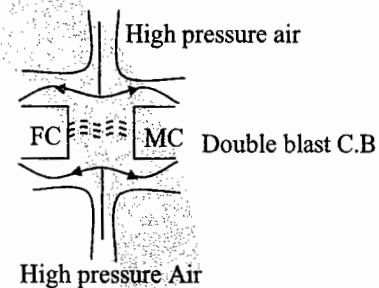
- High pressure air will be exerted on to the arc axially

(ii) Cross blast circuit breaker:



- High pressure air will be exerted on to the arc perpendicularly. So arc resistance will increase and heat energy developed by the arc will reduce. So it will reduce the thermal ionization process.

(iii) Radial blast circuit breaker:



- High pressure air will be exerted on to the arc perpendicularly and It will leave arcing chamber axially.
- This method is used for modern extra high voltage circuit breakers.

7.10.3 Oil Circuit breaker

- Insulating oil or transformer oil is used for Arc quenching medium.
- If the contacts are separated there is an arc struck between the contacts.
- The heat of the arc evaporates the oil and it is decomposed into hydrogen gas at high pressure.
- The volume of gas produced is one thousand times that of oil decomposed.
- An arc energy decomposes oil into 70 % hydrogen, 5 % methane, 22 % acetylene and 3% ethylene
- Arc is in a bubble of gas and surrounded by the oil.

The arc extinction is facilitates mainly by two process.



- (i) Hydrogen gas cools the arc.
- (ii) Hydrogen gas pushes the oil in between the contacts and then removes the arcing products from the Arc.

Advantages:

- It absorbs the arc energy to decompose the oil into gases which is an excellent cooling property
- It acts as an Insulator and permits smaller clearance between live conductors and earthed components.

Disadvantages:

- It is inflammable and there is a risk of Fire.
- Possible to form an explosive mixture with air.
- Arcing products (carbon particles) remains in the oil. These carbon particles increases with number of operation which will deteriorate the oil, so frequent maintenance of oil is required.

According to the amount of oil used in oil circuit breaker there are two types of circuit breaker.

1. Bulk oil circuit breaker
2. Minimum oil circuit breaker

Purpose	Bulk oil CB	Minimum oil CB
Arc quenching	Oil	Oil
Insulation provision with respect to live parts in the system	Oil	Solid insulating material (porcelain)
Cooling the arc	Oil	Gaseous coolant

- In bulk oil circuit breaker, the contacts are placed inside a rigid mechanical chamber
- Whenever, the contacts are about to separate, a small amount of oil will be sent to the space of contacts, so that, the oil will be converted into gas which will form a gas bubble around the Arc to cool the Arc, or for heat dissipation, gaseous coolant will be sent on to gas bubble

7.10.4. SF₆ gas circuit breaker :

- SF₆ gas is used as arc quenching medium
- Dielectric strength of SF₆ gas is 2 – 3 times of air dielectric strength.

Dielectric strength of SF₆ gas = 80 kV/cm

$$\epsilon_r = 1.0$$

- SF₆ gas is a good coolant and it is having electronegative property.
- Electronegative property is a property of material which will attract electrons or electron density under the presence of SF₆ gas. The electrons from the arc will be removed so that the arc strength gets reduced.
- By using cooling property of SF₆ gas, the heat will be dissipated
- At current zero point, a turbulence force, will be created, inside the circuit breaker chamber, such that the heat dissipation will become more effective, there by the arc will be interrupted at current zero point.
- Operating voltage levels 6.6 kV to 800 kV
- For all voltage levels, this SF₆ gas circuit breaker will be used

7.10.5. Vacuum circuit breaker

- Ideally vacuum means zero pressure (or) '0' torr (or) '0' mm of Hg pressure
- Practically high vacuum is created at 10⁻⁵ torr
- Dielectric strength of vacuum = 10⁷ V/cm
= 10000 kV/cm
- As a dielectric strength of vacuum is very very high

The spacing require between the contacts is very low and it is about mm.

- There is no gaseous atoms or molecules presents in vacuum.
- When ever the contacts are about to separate a small amount of metal vapour will be released from contact bodies, due to mechanical friction between the contacts
- On the metal vapour there are two forces
 - (i) electromagnetic forces due to current flow through contacts
 - (ii) Gravitational force (mg)



- At contact separation time, the electromagnetic forces will dominate the gravitational force, such that the metal vapour will be stayed between the contacts and it forms a conducting path between the contacts. This conducting path is called as plasma.
- At current zero point, the electromagnetic forces become zero and the metal vapour will reach to the bottom of circuit breaker chamber due to gravitational forces. There is no a gaseous atom or metal vapour are available for arc restriking. So arc will be interrupted at first current zero itself.

Operating voltage levels up to 66 kV. For above 66 kV levels the creation of vacuum is difficult as well as it is costly.

7.11. Comparison Of circuit Breaker:

1. Insulating Methods.

Property	Air Breakers	Oil	SF ₆	Vacuum
Number of operations	Medium	Low	Medium	High
'soft' break ability	Good	Good	Good	Fair
Monitoring of Medium	N/A	Manual Test	Automatic	Not possible
Fire hazard risk	None	High	None	None
Health Hazard risk	None	Low	Low	None
Economical voltage	Up to 1 kV	3.3-2.2 kV	3.3-800kV	3.3-36 kV

Type	Arc Quenching Medium	Voltage Range and Breaking Capacity
Miniature Circuit Breakers	Air at atmospheric Pressure	400-600V; for Small current taking rating
Air Break Circuit Breakers	Air at atmosphere pressure	400 V- 11 kV; 5-750 MVA
Minimum Oil Circuit Breakers	Transformer Oil	3.3 kV – 220 kV; 150-25000 MVA
Vacuum Circuit Breakers	Vacuums	3.3- 765 kV; 1000-50,000 MVA
SF ₆ circuit Breakers	SF ₆ at 5 kg/cm ² pressure	3.3-765 kV; 1000-50,000 MVA
Air Blast Circuit Breakers	Compressed air at high pressure (20-30 kg/cm ²)	66 kV – 1100 kV; 2500-60,000 MVA ;

Rated Voltage	Choice of circuit Breakers	Remark
Below 1 kV	Air break C.B vacuum C.B., SF ₆ C.B,	Vacuum is preferred
132 kV – 220 kV	Minimum oil C.B SF ₆ C.B air blast C.B,	SF ₆ is preferred
400 kV- 760kV	SF ₆ C.B., air Blast C.B	SF ₆ is preferred.



2. Features of CB:

Factor	Oil Breakers	Air Breakers	Vaccum/SF6
Safety	Risk of explosion and Fire due to increase in pressure during multiple operations	Emission of hot air and ionized gas to the surroundings	No risk of explosion
Size	Quite large	Medium	Smaller
Maintenance	Regular oil replacement	Replacement of arcing contacts	Minimum lubrication for control devices
Environmental Factors	Humidity and dust in the atmosphere can change the internal properties and affect the dielectric		Since sealed, no effect due to environment
Endurance	Below Average	Average	Excellent

Example 7.1:

An LL fault occurs at the terminals of a 3- ϕ , 11kV synchronous generator which is solidly grounded. The fault current lags the system voltage by 60° and voltage of the system is decreased by 5% due to demagnetizing effect of synchronous machine. Calculate active recovery voltage.

Solution:

$$V_{arv} = K_1 K_2 V_m \sin \phi$$

$$\phi = 60^\circ$$

$$K_1 = 0.95$$

(Due to demagnetizing voltage is decreased by 5%)

$$K_2 = 1.5 \text{ (LL fault)}$$

$$V_{arv} = K_1 K_2 V_m \sin \phi$$

$$= 0.95 \times 1.5 \times \frac{11\text{kV}}{\sqrt{3}} \times \sqrt{2} \times \sin 60^\circ$$

$$= 11.08 \text{ kV.}$$

Example 7.2:

A 3- ϕ , 33 kV oil circuit breaker is rated for 1200A 200 MVA and 4 sec

- The symmetrical breaking current is
- Its making current will be

Solution:

- Symmetrical breaking capacity (SBC) = 2000 MVA

$$V_{LL} = 33 \text{ kV}$$

$$\text{S.B.C} = \sqrt{3} \times V_{LL} \times I_{sy}$$

$$I_{sy} = \frac{2000 \times 10^6}{\sqrt{3} \times 33 \times 10^3}$$

$$= 35 \text{ kA}$$

- Making current = $2.55 \times$ symmetrical breaking current

$$= 2.55 \times 35 \text{ kA}$$

$$= 89.22 \text{ kA}$$

Example 7.3:

A 3- ϕ fault on a 33 kV, 50Hz system will give a steady state fault current of 2000A. A 33 kV C.B is placed for this interruption. The phase ground capacitance and inductance up to C.B location is $0.01 \mu\text{F}$ and 5 mH respectively. If arc is chopped at peak value of fault current then voltage across CB contacts is

Solution:

$$\text{fault current} = 2000 \text{ A}$$

$$\text{Maximum value of fault current} = 2000 \times \sqrt{2}$$

$$v = i \sqrt{\frac{L}{C}}$$

$$= 2000 \times \sqrt{2} \times \sqrt{\frac{5 \times 10^{-3}}{0.01 \times 10^{-6}}} = 2 \text{ MV}$$



Example 7.4:

In a 220 kV system, the reactance and capacitance up to the location of circuit breaker is 8Ω and $0.025 \mu\text{F}$, respectively. A resistance of 600 ohms is connected across the contacts of the circuit breaker. Determine the following.

Natural Frequency of oscillation.

Damped Frequency of oscillation.

(iii) Critical value of resistance which will give no transient oscillation.

(iv) The value of resistance which will give damped frequency of oscillation, one fourth of the natural frequency of oscillation.

Solution:

$$L = \frac{8}{2\pi 50} = \frac{8}{100\pi} = 0.02544 \text{ H}$$

$$\begin{aligned} \text{(i) Natural Frequency of oscillation} &= \frac{1}{2\pi} \sqrt{\frac{1}{LC}} \\ &= \frac{1}{2\pi} \sqrt{\frac{1}{0.02544 \times 0.025 \times 10^{-6}}} = 6.304 \text{ kHz} \end{aligned}$$

(ii) Frequency of damped oscillation is given by

$$\begin{aligned} f &= \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{1}{4C^2 R^2}} \\ &= \frac{1}{2\pi} \sqrt{\frac{1}{0.02544 \times 0.025 \times 10^{-6}} - \frac{1}{4(0.025 \times 10^{-6})^2 \times (600)^2}} \\ &= \frac{1}{2\pi} \sqrt{\frac{10^{10}}{6.36} - \frac{10^{10}}{9}} = 3.413 \text{ kHz} \end{aligned}$$

(iii) The value of critical resistance

$$R = \frac{1}{2} \sqrt{\frac{L}{C}} = \frac{1}{2} \sqrt{\frac{0.02544}{0.025 \times 10^{-6}}} = 504.35 \Omega$$

(iv) The damped frequency of oscillation is

$$\frac{1}{4} \times 6.304 \text{ kHz} = 1576 \text{ Hz}$$

$$1576 = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{1}{4C^2 R^2}}$$

$$\frac{1}{2\pi} \sqrt{\frac{1}{0.02544 \times 0.025 \times 10^{-6}} - \frac{1}{4(0.025 \times 10^{-6})^2 \times R^2}}$$

$$\text{or } 1576 = \frac{1}{2\pi} \sqrt{\frac{10^6}{6.36} - 10^{16} 25 R^2}$$

$$\therefore R = 520.8 \Omega$$

Example 7.5:

For a 132 kV system, the reactance and capacitance up to the location of the circuit breaker is 3 ohms and $0.015 \mu\text{F}$, respectively. Calculate the following.

(i) The frequency of transient oscillation.

(ii) The maximum value of restriking voltage across the contacts of the circuit breaker.

(iii) The maximum value of RRRV

Solution:

(i) The frequency of transient oscillation.

$$\begin{aligned} L &= \frac{3}{2\pi 50}, f = 50, \text{ the system frequency} \\ &= \frac{3}{100\pi} = 0.00954 \text{ H} \end{aligned}$$

$$\begin{aligned} f_n &= \frac{1}{2\pi \sqrt{LC}} \\ &= \frac{1}{2\pi \sqrt{0.00954 \times 0.015 \times 10^{-6}}} \\ &= \frac{10^5}{2\pi \times 1.1962} = \frac{10^5}{7.5241} = 13.291 \text{ kHz} \end{aligned}$$

(i) The restriking Voltage

$$V_C = E[1 - \cos \omega_n t]$$

The maximum value of the restriking voltage = $2 E_{\text{peak}}$

$$= 2 \times \frac{132}{\sqrt{3}} \sqrt{2} = 215.56 \text{ kV}$$

(iii) The maximum value of RRRV = $\omega_n E_{\text{peak}}$

$$\begin{aligned} &= 2\pi f_n \times \frac{132}{\sqrt{3}} \times \sqrt{2} \text{ kV/s} \\ &= 9.01045 \text{ kV}/\mu\text{s} \end{aligned}$$